12 kW Wind Turbine for Iceland: Generator Rev C, Complete Design (Rotor and Stator) with One N45SH 30 × 15 × 12 Block per Pole

Status: **design for review** · 2026-11-05 (Martin: “Ok let’s keep this design double check everything and update the whole generator design with 30x15x12 N45SH (one block) create wholistic document with the rotor and stator, complete drawings and isometric views ask me if something unclear or you suggest something”)
Replaces for building: rotor of 19 – Magnet geometry Rev B and 20 – N42SH variant; stator of 18 – Manufacturing §5.1 (unchanged, drawn again here)
Builds on: 06 – Generator round 2 (electromagnetic design, thermal model), 18 (segmented stator, process), 21 – Optimal N42SH shape, 22 – Smaller drum (Ø646 kept)
Geometry: `cad/nacelle_p1/geometry.py` (master, now Rev C) and `cad/generator_rc/rotor_rc.py`
Drawings: `cad/generator_rc/` → `out/NP1-generator-RevC.pdf` (11 sheets), `out/dxf/` (CAD), `out/iso/` (isometric views)
Analysis: `analysis/generator/sh30.py` `S1x` → `sh30_S1x.json`; `checks_rc.py` → `rc_checks.json`; `passive_cooling.py` → `passive_cooling.json`

Summary

Rev C is the doc 20 rotor with each bonded pair of 6 mm blocks replaced by **one solid N45SH block, 30 × 15 × 12 mm, magnetised through the 12 mm**. The drum stays at Ø646. The stator is the doc 18 segmented stator, and its magnetic shape and 28-turn winding are unchanged. Everything was checked again: the new FEA, demagnetisation, temperatures, bond and drum strength, unbalanced pull and voltage.

	Rev B (doc 19)	Doc 20 (N42SH pairs)	Rev C (this document)
Magnet	N40UH/EH 40 × 20 × 7.5	N42SH 30 × 15 × 6, bonded pairs	**N45SH 30 × 15 × 12, one block**
Pieces per generator	160	400 (200 pairs)	**200 + 10 spare**
Magnet mass	8.0 kg	8.2 kg	**8.2 kg**
Drum seat	Ø618.65	Ø628.05	**Ø628.05** (unchanged from doc 20)
Line EMF, 178 rpm, 100 °C	380 V (28 turns)	378 V (29 turns)	**372.0 V** (28 turns)
Torque at the sizing current density (2D, 100 °C, doc 22 method)	n/a	760 Nm	**772.6 Nm** (+1.7 %)
Cogging, skewed	small	≈ 0	**0.01 Nm** pk-pk (5.4 Nm unskewed)
Transient short-circuit margin at 100 °C	see doc 19	32 %	**30.1 %** (rule ≥ 20 %)
Turns per coil	28	29	**28**
Bonding steps	1	2 (pair, then seat)	**1**

What changed against doc 20: no pair bonding step (half the pieces, no glue line inside the magnet), slightly stronger grade, 28 turns kept instead of 29. **What did not change:** drum, seat,

comb bars, end rings, sleeve, stator, coils, carrier and the doc 18 process.

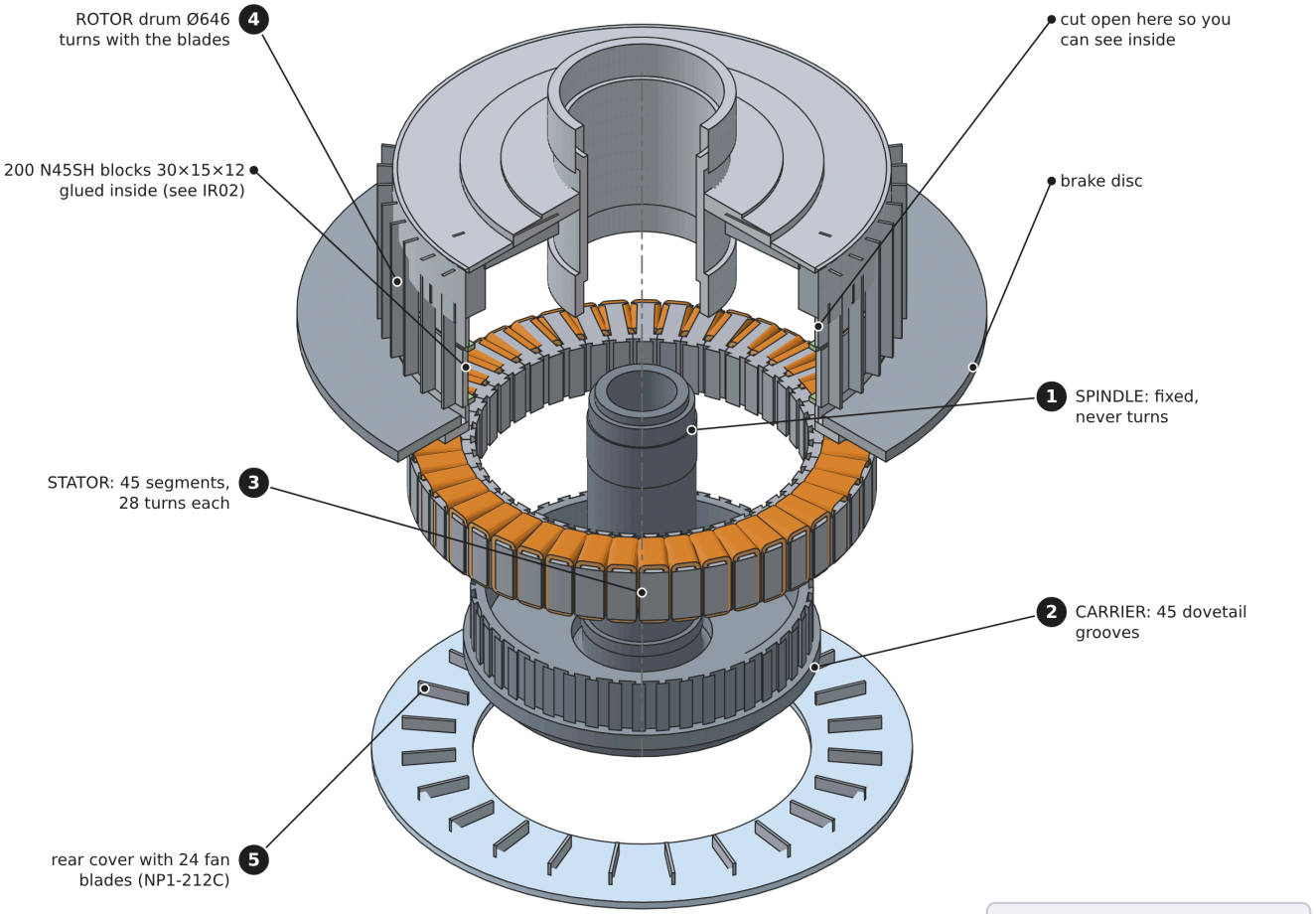
Results of the double check:

- **Electromagnetic: OK.** 372 V at 100 °C with 28 turns; the controller works the same as with Rev B. 1.7 % more torque per ampere-turn than doc 20.
- **Demagnetisation: OK.** 30.1 % transient short-circuit margin at 100 °C, 22 % at 110 °C. The magnets run at 67 to 72 °C in the worst hot case, where the margin is about 45 %.
- **Thermal: one finding, now fixed with passive cooling.** The doc 06 cooling package (fan blades, inner fins, outer ribs) was assumed in the thermal sign-off but never drawn in P1. Without it the winding hot spot reaches 144 to 168 °C on a 30 °C day, above the 130 °C design limit. Rev C now has a **purely passive** package that turns with the rotor: 24 blades on the rear cover, 36 inner fins and 48 outer ribs on the drum. Hot spot **94 to 105 °C**, magnets 55 to 59 °C (§4, §4a).
- **Mechanical: OK, with one condition.** Bond shear is 28 kPa at rated torque and 170 kPa at a short circuit, against ≥ 5 MPa. The drum is stable only because both ends are held round: the front disc and the rear brake-disc flange must be continuous, full-round welds (§5).
- **Voltage: OK.** Highest line peak 852 V (−30 °C, 250 rpm overspeed) against 1,200 V.

Questions and suggestions for Martin are in §10. Martin chose rotor fins and passive cooling on 2026-11-05 and accepted Ø676 over the outer ribs.

IR01 Generator Rev C pulled apart (isometric, exploded)

Front (hub) end up. Parts go together straight down the middle line, in the order 1 to 5.



Order: 1 spindle on the stand, 2 carrier bolted on, 3 slide segments into the grooves, 4 rotor lowered on (J40), 5 rear cover, blades inward.

1. Machine data

Item	Value
Type	Outer-rotor radial-flux PMSG, direct drive, 40 poles / 45 slots, tooth-coil double-layer winding, star
Rated point	802 Nm at 178 rpm, 13.6 kW shaft, 12 kW net to the grid; overspeed limit 250 rpm
Air gap	Ø600 at the magnet face, 2.0 mm mechanical (stator Ø598, sleeve bore Ø602)
Stack	74.0 mm, M270-35A, 0.35 mm self-bonding sheets, 45 tooth segments of 8°
Winding	45 coils, 28 turns, 4 strands Ø1.55 in hand, 15 coils per phase in series, class 200 wire, class H system, VPI + potted ends
Rotor	S355J2+N drum, Ø646 outside, seat Ø628.05 × 90 long (x 335 to 425), 200 blocks in 5 rows of 40
Magnet	N45SH, 30 (across) × 15 (along) × 12 (radial), NiCuNi, 41.0 g
Pole	1 block across, 5 rows along, rows skewed 0.20° (−0.4° to +0.4°), span 5.70° (α 0.633)
Encapsulation	Glass-epoxy sleeve, 0.52 mm at the block centre to 0.89 mm at the edges
Cooling	Passive only: 24 blades on the rear cover, 36 inner fins, 48 outer ribs (no motor, no power)
Masses	Rotor ≈ 161 kg (drum 91.8, fins and ribs 12.5, brake disc 34.5, covers 10, magnets 8.2, G11 2); stator ≈ 115 kg (carrier 40.6, core 48.0, winding 24, clamp ring 2.2)

2. Design basis

- **Magnets are designed at 100 °C**, as doc 19 §16.3: the controller derates at 85 °C and stops at 95 °C magnet temperature (estimated from the EMF).
- **Demagnetisation rule:** after a 3-phase terminal short circuit from rated load, at least 20 % of every block’s volume stays above the knee margin at 100 °C (doc 19).
- **Thermal rule:** winding hot spot ≤ 130 °C at rated load on a 30 °C day (class H system, 180 °C, keeps 50 K reserve for life).
- **Same stator as doc 18.** The tooth shape, slot and yoke are frozen; only the rotor magnet pack changes.

3. Electromagnetic results (FEA, shape S1x)

2D FEA with the doc 06 model (`sh30.py S1x`), N45SH data Br 1.32 T, HcJ 1,592 kA/m at 20 °C, αBr −0.12 %/K, βHcJ −0.60 %/K.

	20 °C	60 °C	100 °C
Line EMF at 178 rpm, 28 turns (skewed)	404.6 V	388.7 V	372.0 V

Magnet temperature	60 °C	80 °C	100 °C	110 °C	120 °C
HcJ (kA/m)	1,210	1,019	828	732	637
Margin at rated load (%)	68.2	63.0	55.4	50.2	43.3
Margin, steady short circuit (%)	60.5	54.3	45.3	39.0	30.8
Margin, transient short circuit (%)	49.7	41.7	**30.1**	22.0	11.4
Peak short-circuit current (A)	66.3	64.4	62.6	61.7	60.8

- **Turns:** 380 V at 100 °C would need 28.6 turns. 28 turns are kept: the controller and doc 12 converter work over the whole EMF range, the slot layer plan (37/37/31/7 of 125 positions) stays as doc 18, and copper loss does not depend on the turn count for the same slot fill. Rated torque needs 380/372 = 2.2 % more current than the doc 06 basis, so the copper loss is **518 W** (doc 06 / Rev B 496 W, +4.3 %; doc 20 with 29 turns was +8 %).
- **Torque:** 772.6 Nm (2D) at the sizing current density, against 760 Nm for doc 20. With the same 3D correction as doc 22, about 815 Nm, so the 802 Nm rated torque is reached with 1.6 % less current than the basis.
- **Cogging:** 5.35 Nm pk-pk with straight rows, 0.01 Nm with the 0.20° step skew.
- **Rotor eddy loss:** 4.7 W in the solid blocks (S1d result, same block width), negligible.
- **Annual energy:** the extra 22 W of copper loss at rated is under 0.1 % of the annual energy (estimate from doc 22: 31.6 MWh at 6 m/s, 51.1 MWh at 8 m/s).

4. Thermal check and the cooling finding

The doc 06 lumped thermal model (`thermal.py`) was run with the Rev C losses (copper 518 W, iron 218 W) and the P1 geometry (stack 74 mm, slot 65 mm, drum Ø646 × 194). The doc 06 sign-off used an “improved” cooling package: fan blades on the rotor end cover (inside air h 40 W/m²K), inner drum fins (3 × area), outer ribs (1.5 × area). **None of these are in the P1 drawings.**

Winding hot spot / magnet temperature (°C), ambient 30 °C, 100 W sun:

Cooling	Rated, 10.5 m/s	Rated, weak wake	110 % load	Iceland 10 °C, rated
P1 as drawn before Rev C (no fan)	**144.1** / 66	**148.9** / 71	**167.7** / 70	113.4 / 39
Rev C as drawn: 24 fan blades on the rear cover	108.8 / 67	114.0 / 72	122.1 / 71	80.0 / 40
Rev C passive package (blades + inner fins + outer ribs), §4a	**94.0** / 55	**97.7** / 59	**104.6** / 58	67.3 / 31
Doc 06 assumption, for reference (3 × inner area, 1.5 × outer)	93.0 / 56	97.0 / 60	103.1 / 59	66.2 / 32

Design limit 130 °C hot spot, 100 °C magnet.

- **Fan blades** (NP1-212C): 24 blades 4 mm × 18 mm high, r 240 to 308, welded on the inside of the rear cover at x 447 to 465. They stir the air between end windings and drum, 4 mm clear of the drum bore (Ø624).
- **Inner fins and outer ribs:** see §4a.
- Magnet temperatures stay at 55 to 59 °C with the passive package (67 to 72 °C with blades only), well inside the 100 °C design point.

4a. Passive cooling study (Martin, 2026-11-05: “I think the rotor deserves fins, can we explore purely passive cooling”)

Everything here is passive. There is no electric fan, pump or power draw. The parts are welded to the rotor, and they move air only because the rotor turns. The 24 “fan blades” on the rear cover are themselves just radial fins on a part that turns. Script: `passive_cooling.py` → `passive_cooling.json`.

Parts (now on NP1-210C and NP1-G00):

Part	Geometry	Added area	Fin efficiency	Mass
36 inner fins	L-shaped 3 mm S355 plates every 10°, welded to the hub web and the shell bore in the front cavity: x 220 to 312, r 262 to 312, plus a web leg x 220 to 250, r 140 to 262. 9 mm clear of the end windings (x 321)	0.60 m² (+66 % of the inside area 0.90 m²)	0.78	7.0 kg
48 outer ribs	4 × 20 S355 flat bars every 7.5° on the shell outside, x 222 to 430, tips Ø676, clear of the brake disc (x 435)	0.39 m² (+54 % of the outside area 0.72 m²)	0.95	5.5 kg
24 cover blades	as §4 (NP1-212C)	small		in the cover

The doc 06 assumption of 3 × the inside area cannot be reached in the real cavity; the drawn fins give about 1.5 ×. So the earlier “inner fins give 7 to 9 K” in this document’s first issue was too optimistic: they give 3 to 4 K on their own. The outer ribs are the bigger passive gain.

Winding hot spot / magnet temperature (°C), 30 °C ambient unless stated:

Passive option	Rated	Weak wake	110 % load	Iceland 10 °C
A. No cooling parts	144.1 / 66	148.9 / 71	167.7 / 70	113.4 / 39
B. Cover blades only (Rev C first issue)	108.8 / 67	114.0 / 72	122.1 / 71	80.0 / 40
C. Inner fins only, air barely stirred (h 15)	139.1 / 66	143.9 / 71	161.1 / 70	108.6 / 39
C’. Inner fins only, fins stir the air (h 25)	120.4 / 66	125.4 / 72	136.8 / 71	90.9 / 40
D. Outer ribs only	133.6 / 54	136.9 / 58	155.0 / 57	104.7 / 30
E. Inner fins + outer ribs, no blades (h 25)	109.2 / 55	112.8 / 58	123.7 / 58	81.6 / 30
F. Blades + inner fins	105.5 / 67	110.7 / 72	117.9 / 71	76.9 / 40
G. Blades + inner fins + outer ribs (chosen)	**94.0** / 55	**97.7** / 59	**104.6** / 58	67.3 / 31

Passive option	Rated	Weak wake	110 % load	Iceland 10 °C
G with the inside air only half as stirred (h 25)	108.8 / 55	112.4 / 58	123.2 / 58	81.2 / 30
H. G with 25 mm ribs (tips Ø696)	89.2 / 50	92.2 / 53	99.1 / 52	63.3 / 26

What the numbers say:

- **Moving the inside air is the biggest lever, more than fin area.** Heat leaves the coils into the inside air and the stator core. Fins on the drum only help if that air is moving over the end windings. Inner fins in still air (C) gain only 5 K. Inner fins alone are not enough (C’, 137 °C at 110 %).
- **Outer ribs cool the magnets and the drum** (11 to 13 K lower magnet temperature) and take 10 to 14 K off the hot spot.
- **G is the recommended passive package:** 94 to 105 °C on the hot cases, 25 K under the 130 °C limit. Even if the inside air moves only half as well as doc 06 assumed, it stays at 123 °C or lower (row “G with h 25”). The magnets stay below 60 °C, where the demag margin is about 50 %.
- **H (taller ribs)** gains another 5 K. It is only worth it if the bench test shows the hot spot above about 110 °C, and only if the nacelle cover has room.

Costs and checks:

- 12.5 kg more on the rotor (161 kg). Welded before stress relief and machining, so the drum is machined and balanced with them.
- **Ø676 over the ribs:** accepted by Martin (2026-11-05). Confirm the caliper and cover-lip clearance in the next nacelle model update (docs 07, 16).
- Rib windage at 178 rpm is a few watts. The ribs are 4 mm flat bars with rounded ends, so they whistle less than thin blades.
- Paint the ribs as the drum (zinc spray + epoxy, C5). Keep the inside fins bare or thinly primed: paint is a thermal barrier.

5. Mechanical checks

Check	Value	Limit / comment
Bond area per block	450 mm² (30 × 15 on the seat)	
Bond shear at rated torque	28 kPa	Epoxy ≥ 5 MPa at 120 °C
Bond shear, transient short circuit (6 × rated)	170 kPa	Margin 29
Centrifugal force per block at 250 rpm	8.6 N outward	
Magnetic pull per block at rated	30 N outward	Bond in **compression**: both forces push the block into the seat
Drum wall under the seat	8.98 mm	
Net magnetic pressure on the drum	44 kPa	
Hoop stress (magnetic + magnets + own spin at 250 rpm)	2.5 MPa	S355 yield 355 MPa
Unbalanced magnetic pull	1.29 kN per mm of eccentricity	Doc 18: 1.9 (thinner magnet); J40 tooling OK
Drum stability, ring only (ends free)	stiffness 1.1e7 against magnetic 1.85e7 Pa/m: **0.6**	Not enough on its own
Drum stability, both ends held round (Donnell shell, 239 mm)	5.6e9 against 1.85e7 Pa/m: **302**	OK

Condition from the stability check: the 9 mm shell is safe only because the front disc and the rear brake-disc flange keep its ends round. Drawing NP1-210C now asks for full-round continuous welds at both and a roundness check of the seat (0.05) after stress relief. A drum with a free rear edge would not be acceptable.

6. Voltage and insulation

	Value
Line EMF at −30 °C, 178 rpm	428.9 V rms
Highest line peak, −30 °C, 250 rpm overspeed	852 V
Limit (converter 1,200 V class, winding test 2.5 kV)	1,200 V

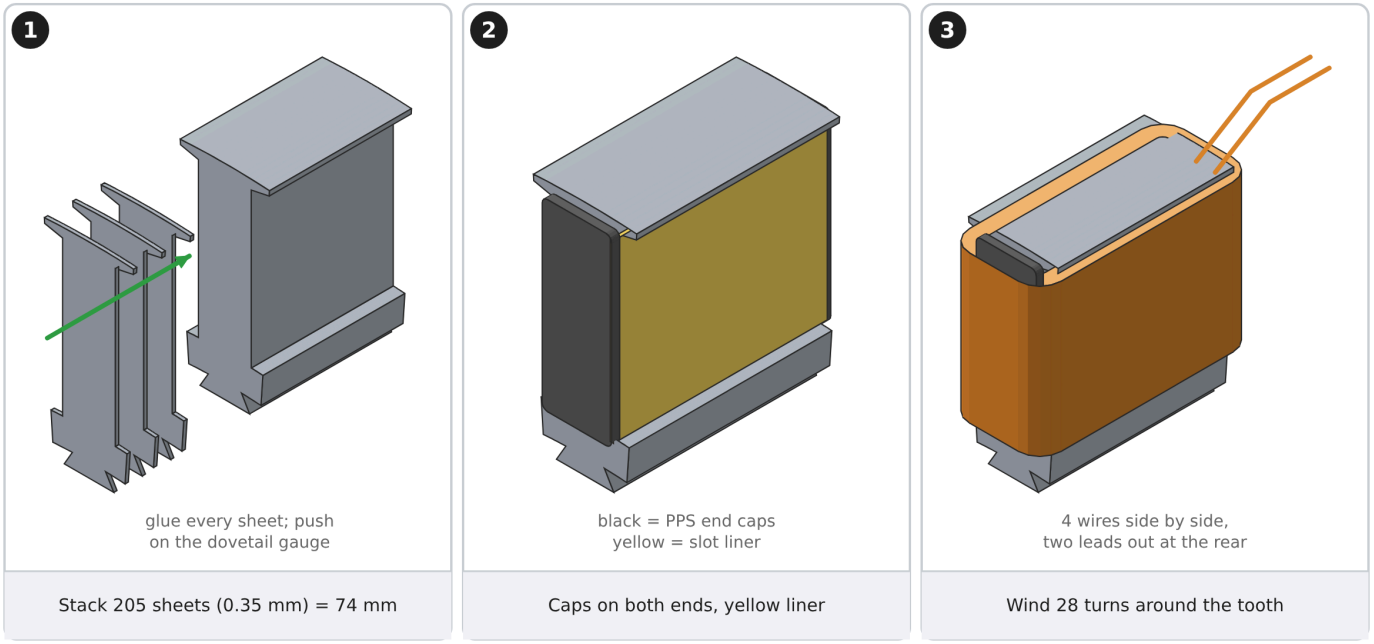
7. Stator (unchanged from doc 18, drawn again)

The stator is the doc 18 segmented stator. The Rev C sheets redraw it with the same numbers so the set is complete.

- **Segment NP1-201C:** one tooth per segment, 8° pitch, 205 sheets M270-35A bonded to 74.0 mm. Dovetail foot 14 wide at r 218.6, 6 deep, 60° flanks (20.92 at the bottom). 45 + 3 spare per stator, 1.07 kg each.
- **Carrier NP1-200C:** S355 ring with 45 dovetail grooves, slide fit, rear stop face at x 417, bolted to the spindle flange.
- **Coil NP1-203C:** 28 turns, 4 × Ø1.55 in hand, layer-wound directly on the insulated tooth (PPS end caps, NKN slot liner), layers 37/37/31/7 of 125 positions, about 0.45 kg copper per coil.
- **Connection:** 15 coils per phase in series, star, coil order and polarity per `coil_connection.csv` (doc 18 §5.9).
- **Clamp ring NP1-202:** 1.4404 stainless, front of the stack.
- **Process:** doc 18 §5 unchanged (bond, inspect, insulate, wind, test, assemble, connect, VPI, pot).

I02 One stator segment in 3D: stack it, insulate it, wind it

Isometric views, drawn 3× larger than real. The tooth points up, the dovetail foot is at the bottom.



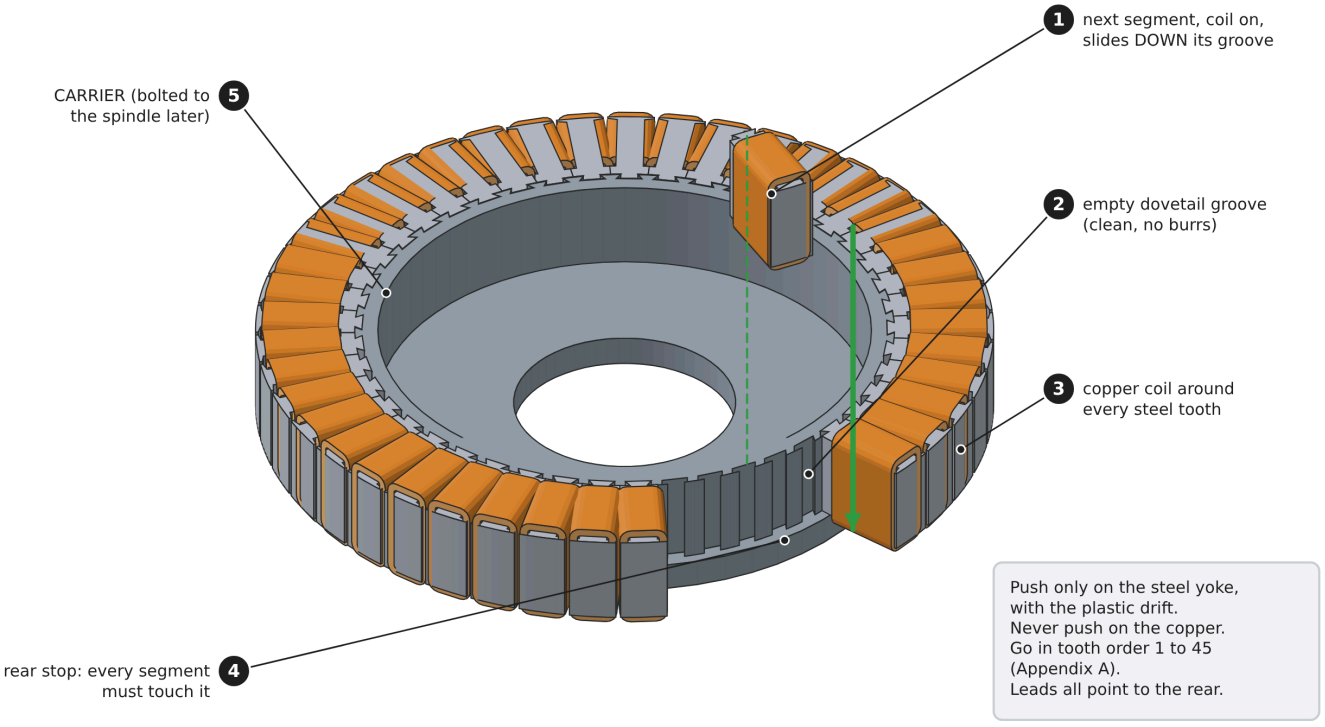
Stack 205 sheets (0.35 mm) = 74 mm

Caps on both ends, yellow liner

Wind 28 turns around the tooth

I03 Slide the wound segments down into the carrier grooves (stand NP1-T05)

Isometric view, front end up. 40 segments are in, the next one goes straight down its groove.

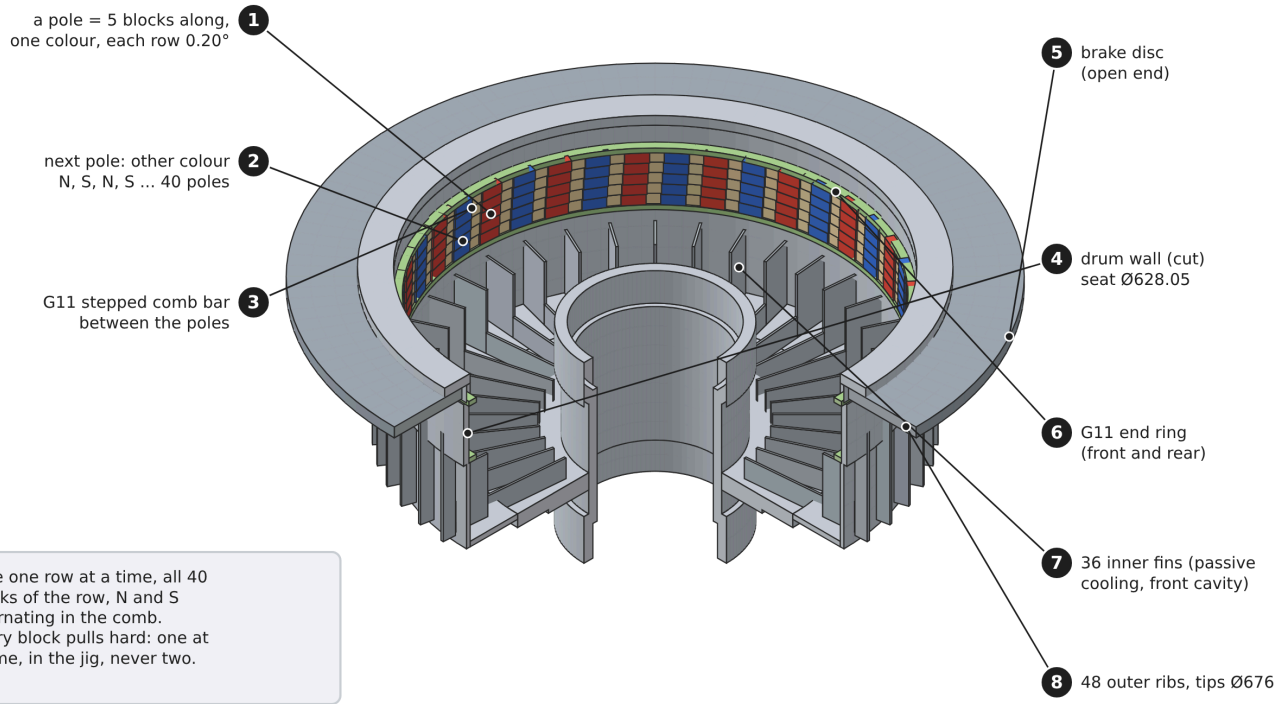


Push only on the steel yoke, with the plastic drift.
Never push on the copper.
Go in tooth order 1 to 45 (Appendix A).
Leads all point to the rear.

8. Rotor

IR02 Rotor drum Rev C with its 200 blocks (isometric, cut open)

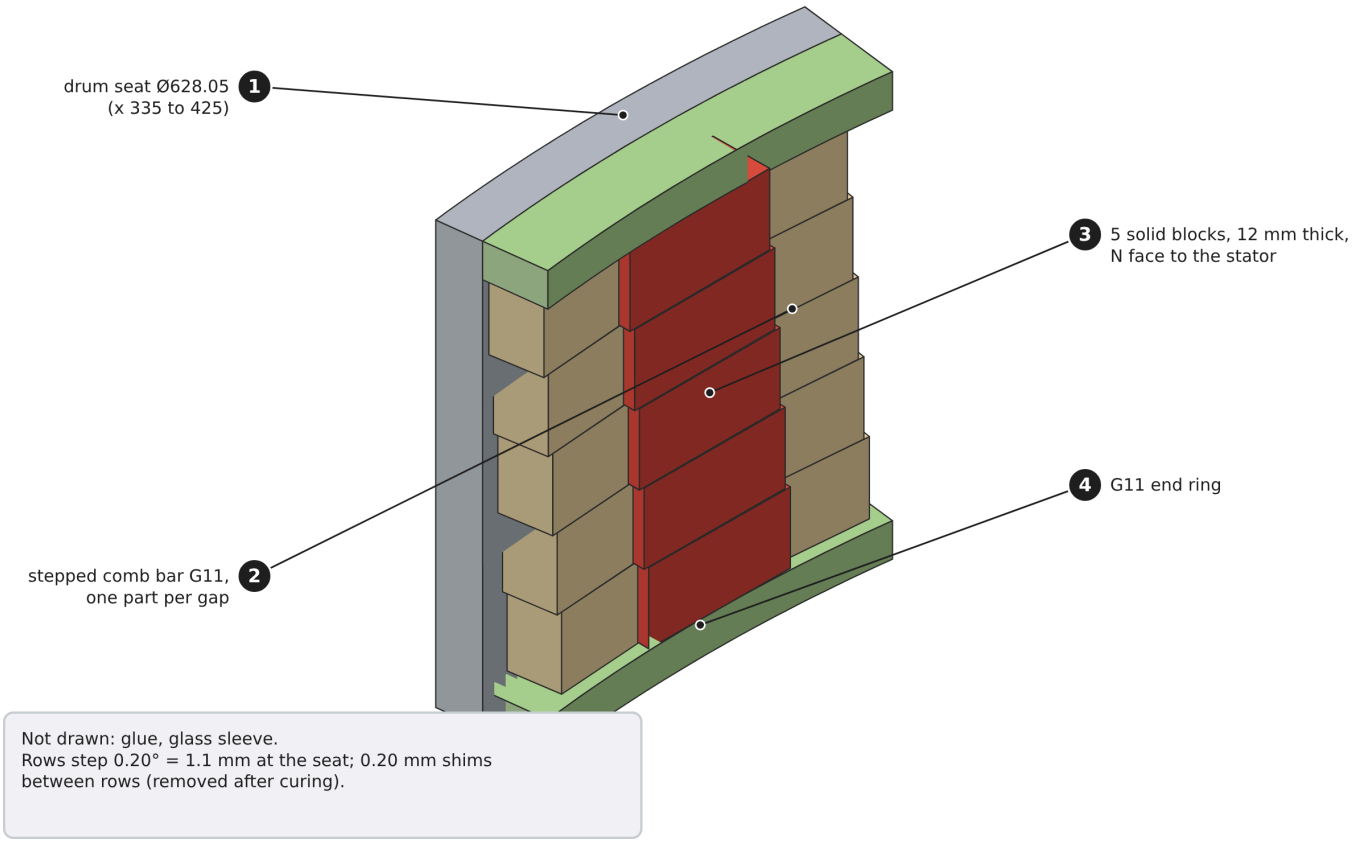
Open (rear) end up, as on the magnet jig. A quarter is cut away so the five skewed rows can be seen.



Glue one row at a time, all 40 blocks of the row, N and S alternating in the comb. Every block pulls hard: one at a time, in the jig, never two.

IR03 One pole of Rev C (isometric), seen from the stator side

Five solid N45SH blocks 30 × 15 × 12 on the drum seat, each row turned 0.20° further. Comb bars either side.



Drum NP1-210C. As doc 20: seat Ø628.05 +0.05/0 over x 335 to 425, 8.98 mm of steel under the seat, outside Ø646. Seat roundness 0.05 and runout 0.05 to the bearing seats after stress relief. Full-round welds at the front disc and at the brake-disc flange (§5).

Magnets NP1-220E (purchase drawing).

Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m, (BH)max ≥ 342 kJ/m³ at 20 °C
Size	30 ±0.05 × 15 ±0.05 × 12 ±0.03; faces flat and parallel within 0.02
Magnetisation	Through the 12 mm, saturated, axis within 2° of the face normal
Edges / coating	Chamfer 0.2 to 0.3; NiCuNi 10 to 20 µm, 72 h salt spray
Uniformity	Moment within ±1.5 % of the lot mean, value per serial
Data	J-H curves per batch at 20, 60, 100, 120 °C
Marking	White dot on the N face
Quantity	200 + 10 spare per generator, + 8 for the Q1 tests

Comb bars NP1-221C. Stepped G11 bar between every two poles (40 + 4 spare), 82 long, 10 high, 17.54 wide at the bottom and 19.11 at the top. Five steps of 1.096 mm, one per row, give the 0.20° skew without a skewed fixture.

End rings NP1-222C. G11, 7.1 wide, from the seat to the block inner face (12.5 radial), front and rear. They close the pack and hold the comb ends.

Magnet pack NP1-223C. Rows at x 342.1, 357.3, 372.5, 387.7, 402.9 (15 mm blocks, 0.20 mm removable shims), 75.8 mm long, 0.9 overhang at each end of the stack. Bond line 0.15 at the corners, 0.51 at the middle (flat block on a round seat). Sleeve 0.52 to 0.89.

Bonding (replaces doc 18 §6.5 and doc 20’s pair step). Place one row at a time with the doc 19 jig: all 40 blocks of the row, N and S alternating, each one pushed sideways into its comb pocket, never let to snap. Shims out after cure, then the next row. Each block goes down once; there is no pair bonding.

Rear cover NP1-212C. 5083 plate with the seal carrier and the 24 fan blades on the inside (§4). Balance the rotor with the cover fitted.

9. Drawings and views

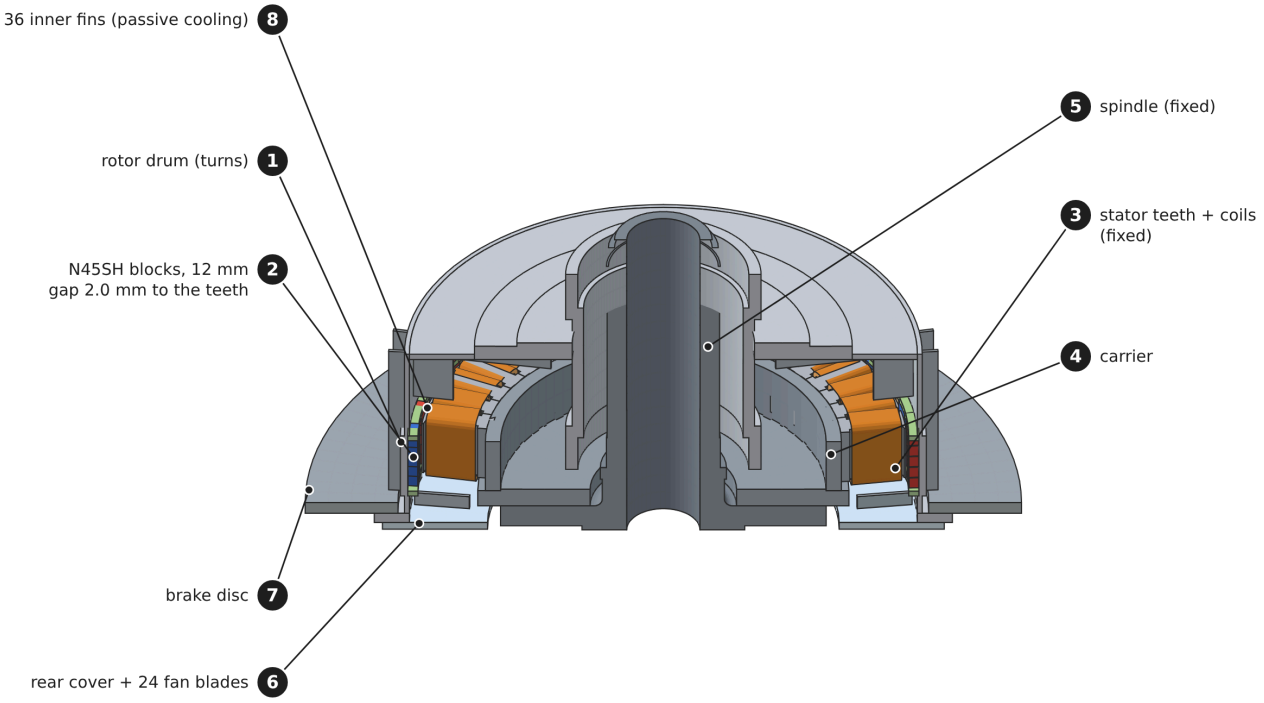
The drawing set is `cad/generator_rc/out/NP1-generator-RevC.pdf`, with a PDF and a DXF of each sheet. Regenerate with `python3 build_rc.py --png`.

Drawing	Title	Material	Mass
NP1-G00	Generator Rev C, assembly section with parts list		≈ 276 kg generator
NP1-201C	Stator tooth segment lamination (45 + 3)	M270-35A, 0.35 mm, self-bonding	1.07 kg per stack
NP1-200C	Stator carrier with 45 dovetail grooves	S355J2+N	40.6 kg
NP1-203C	Coil on segment	Cu Ø1.55 grade 2, class 200; NKN; PPS GF40	≈ 0.45 kg Cu
NP1-202	Stator front clamp ring (unchanged)	1.4404	2.2 kg
NP1-210C	Rotor drum weldment with inner fins and outer ribs	S355J2+N	91.8 + 12.5 kg
NP1-220E	Magnet block 30 × 15 × 12 (purchase drawing)	N45SH, NiCuNi	41.0 g
NP1-221C	Stepped pole comb bar (40 + 4)	G11	≈ 25 g
NP1-222C	Magnet end rings, front and rear	G11	≈ 0.12 kg each
NP1-223C	Rotor magnet pack: pole section, layout, sleeve	assembly	8.2 kg
NP1-212C	Drum rear cover with seal carrier and fan blades	EN AW-5083	≈ 9.0 kg

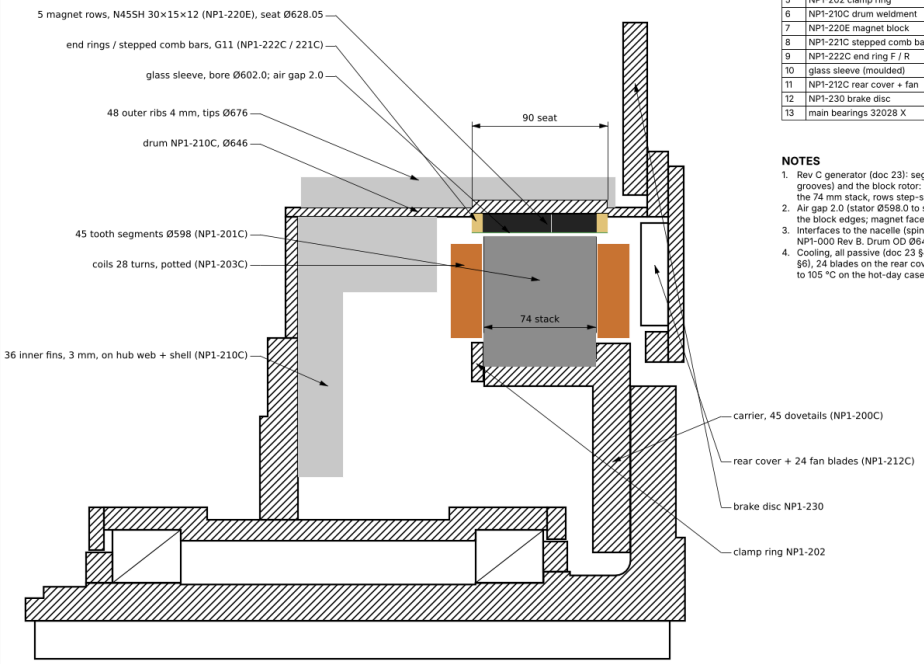
Isometric views (`python3 iso_rc.py --png`): IR01 exploded generator, IR02 rotor with 200 blocks, IR03 one pole, IR04 assembled section, IR05 segment, IR06 segments into the carrier.

IR04 Generator Rev C assembled (isometric, cut in half)

Front end up, the half nearest you cut away. The magnets run 2 mm outside the stator teeth all the way round.



GENERATOR SECTION (upper half, 1:2.2)



#	Part	Qty	Material / note
1	NP1-100 spindle (stationary)	1	42CrMo4+QT
2	NP1-200C stator carrier	1	S355J2+N, 45 dovetails
3	NP1-201C tooth segment stack	45+3	M270-35A bonded, 205 sh.
4	NP1-203C coil on segment	45+3	28 t × 4 × Ø1.55, cl. 200
5	NP1-202 clamp ring	1	1.4404
6	NP1-210C drum weldment	1	S355J2+N, seat Ø628.05
7	NP1-220E magnet block	200+10	N45SH 30×15×12
8	NP1-221C stepped comb bar	40+4	G11
9	NP1-222C end ring F / R	1+1	G11
10	glass sleeve (moulded)	1	E-glass/epoxy 0.5–0.9
11	NP1-212C rear cover + fan	1	5083, 24 blades
12	NP1-230 brake disc	1	C45E+N
13	main bearings 32028 X	2	bought

NOTES

1. Rev C generator (doc 23): segmented stator of doc 18 (45 tooth segments in carrier dovetail grooves) and the block rotor: one N45SH block 30 × 15 × 12 per pole across, five rows along the 74 mm stack, rows step-skewed 0.20° by the stepped comb bars.
2. Air gap 2.0 (stator Ø598.0 to sleeve bore Ø602.0); sleeve 0.52 at the block centre, 0.89 at the block edges; magnet face r 301.52.
3. Interfaces to the nacelle (spindle, bearings, hub flange, disc, bedplate) unchanged from NP1-000 Rev B. Drum OD Ø646 unchanged.
4. Cooling, all passive (doc 23 §4 and §4a): carrier → spindle → finned insert path (doc 14 §6), 24 blades on the rear cover, 36 inner fins and 48 outer ribs on the drum; hot spot 94 to 105 °C on the hot-day cases (limit 130).

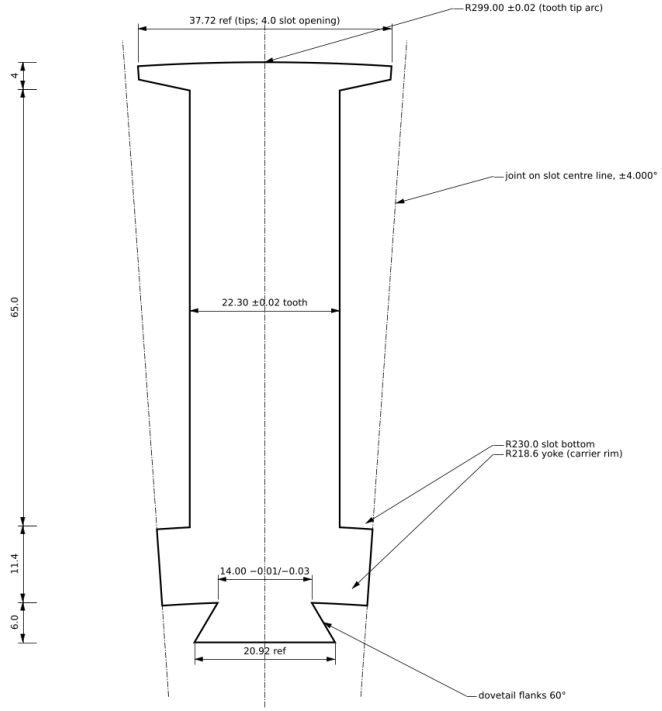
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

GENERATOR REV C · ASSEMBLY SECTION

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-G00	C · 2026-11-05	1:2.2
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
see parts list	≈ 330 kg drivetrain (no hub)	Claude / not checked

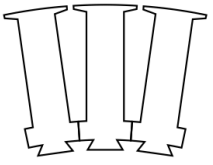
SEGMENT (2:1), radial direction up



NOTES

1. DXF layer OUTLINE = cutting contour. Laser cut (P1), burr ≤ 0.02, heat zone ≤ 0.1; or progressive die (series).
2. Magnetic contour identical to the doc 14 one-piece lamination; joints on the slot centre lines in the yoke (8.000° pitch).
3. Stack: 205 sheets, 74.0 ±0.1 (sheets counted, height measured per segment), bonded in fixture NP1-T03 at 1.0 to 1.5 MPa, varnish cure per supplier.
4. Checks per stack: tip radius R299.00 ±0.02 on gauge, foot width 14.00 −0.01/−0.03, slot opening edges straight 0.05 over 74; Franklin test ≥ 10 Ω·cm².
5. Do not machine tooth tips after stacking. 45 + 3 spare segments per stator; all identical.

3 OF 45 SEGMENTS (1:2)

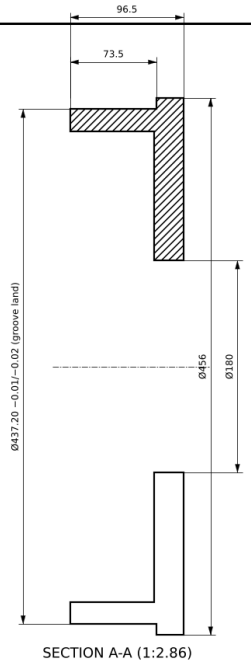


Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

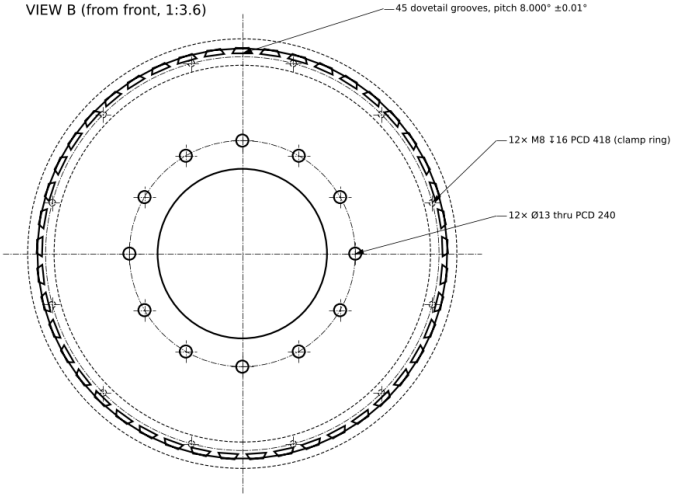
STATOR TOOTH SEGMENT LAMINATION (45 + 3 PER STATOR)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-201C	C · 2026-11-05	2:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
M270-35A, 0.35 mm, self-bonding varnish	1.07 kg per stack	Claude / not checked

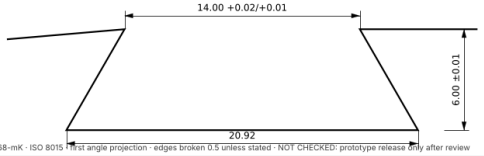


SECTION A-A (1:2.86)

VIEW B (from front, 1:3.6)



DETAIL C: GROOVE (5:1), flanks 60°



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Rev C (doc 18 §5.1): grooves replace the shrink fit and the 3 keys. Grooves open to the front (x 343.5), closed by the rear stack stop (x 417); broached or wire-EDM.
2. Groove angular position ±0.01°, groove seat radius R212.60 ±0.01 to the bolting-face axis; foot clearance 0.01 to 0.03 (slides by hand when clean), fixed by VPI resin, clamped by NP1-202 (12× M8 A4-80, 20 Nm).
3. Run-out of groove lands and rear stop face to the bolting face and Ø180 spigot ≤ 0.02; bolting face flatness 0.02.
4. Bolt to the spindle flange: 12× M12 10.9 + hardened washers, 110 Nm, 2 fitted dowels Ø10 H7 PCD 240; Al-filled paste 0.05 on the full face (heat path, doc 14 §6).
5. Stress relieve 580 to 620 °C, 2 h, before final machining. No paint on grooves and bolting face; elsewhere epoxy primer 60 µm.

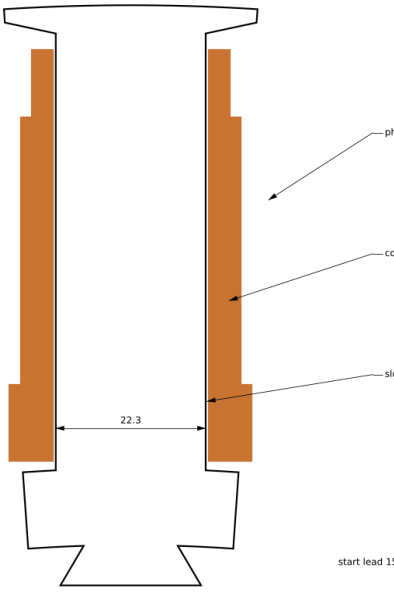
12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR CARRIER WITH 45 DOVETAIL GROOVES

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-200C	C · 2026-11-05	1:2.86 / 1:3.6 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N ring forging or plate, stress relieved	40.6 kg	Claude / not checked

Layer (from tooth flank)	Positions	Used
1	37	37
2	37	37
3	31	31
4	17	7
5	3	0 (reserve)
Total (28 turns × 4 strands)	125	112

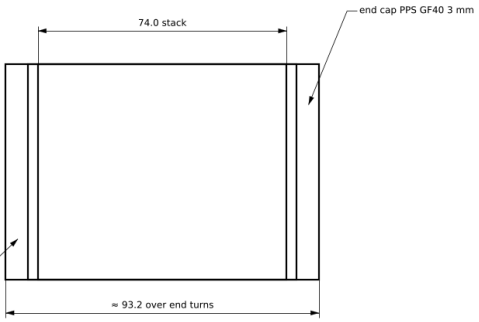
COIL SIDES IN THE SLOT (2:1)



NOTES

1. Wind on chuck NP1-T04 by the dovetail, 4 parallel strands Ø1.55 (≤ 1.66 over enamel), 25 to 30 N per strand, layer by layer from the tooth flank, ≤ 60 rpm (doc 18 §5.6).
2. 28 turns per coil (Rev C keeps doc 18): 373 V line at 178 rpm, magnets 100 °C; 389 V at 60 °C. Wire per coil ≈ 6.6 m per strand; R20 per phase 0.217 Ω.
3. All coils wound the same direction; reversal at connection (doc 18 Appendix A): 15 coils per phase in series, star.
4. Tests before assembly (doc 18 §5.7): resistance ±2 % of the lot mean, surge 3.5 kV, hipot to the segment 2.3 kV 1 min, IR ≥ 1 GΩ.
5. After assembly: VPI class H epoxy, end windings potted (doc 18 §5.11, §5.12). PT100 in 6 separator gaps (2 per phase).

SIDE VIEW OF COIL ON TOOTH (1:1, simplified)



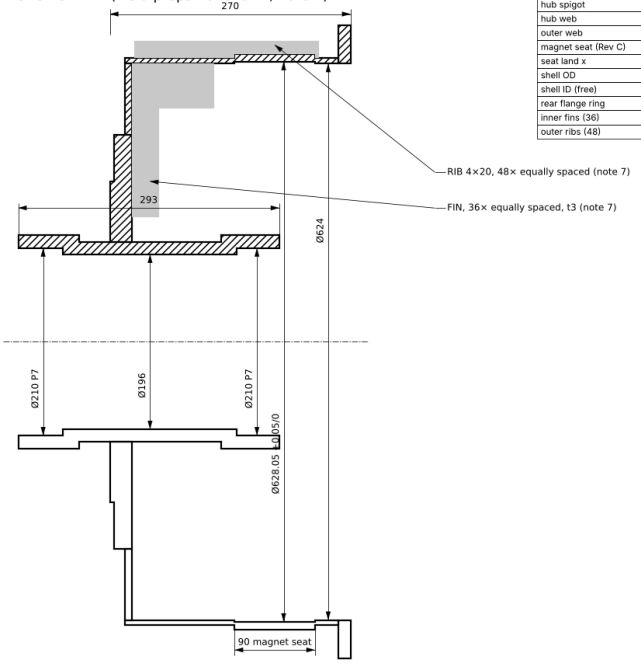
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

COIL ON SEGMENT (WOUND SEGMENT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-203C	C · 2026-11-05	2:1 / 1:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Cu enamelled Ø1.55 grade 2, class 200; NKN 0.30; PPS 45 % GF	5.45 kg Cu per coil	Claude / not checked

SECTION A-A (weld preps not shown, note 2)



Feature	Size	Tolerance / note
cup seats (2)	Ø210 × 45	P7, Ra 0.8, coax. 0.02
cup shoulders	Ø196	square 0.01 to axis
hub spigot	Ø360 × 5	f7; face flat 0.03
hub web	20 thick, Ø464	12× M16 thru PCD 400
outer web	8 thick	full pen. weld to shell
magnet seat (Rev C)	Ø628.05 × 90	+0.05/0, coax. 0.02, Ra 3.2
seat land x	335 to 425	as Rev B
shell OD	Ø646	steel under seat 8.98
shell ID (free)	Ø624	clearance to magnets
rear flange ring	Ø624/Ø710 × 14	M10 PCD 644 + M10 PCD 678
inner fins (36)	t3, x 220-312, r 140-312	L-plate, web + bore, every 10°
outer ribs (48)	4 × 20, x 222-430	tips Ø676, every 7.5°

NOTES

1. Rev C: magnet seat Ø628.05 +0.05/0 (was Ø618.40) for the 12 mm N45SH blocks; steel under the seat 8.98 (FEA: 1.45 T median, 1.7 T at the pole edges, included in the EMF).
2. Welds full penetration at web/housing and web/shell, ISO 5817 B, 100 % MT, UT of the shell seam (ring-rolled shell preferred, doc 18 §4.2). Stress relieve 580 to 620 °C before machining.
3. Machine in one set-up: cup seats Ø210 P7, magnet seat, hub spigot Ø360 f7, rear flange face; coaxiality ≤ 0.02, shoulders square ≤ 0.01.
4. Hub web 12× M16 thru PCD 400; rear flange ring 24× M10 PCD 644 (rear cover) and 24× M10 PCD 678 (disc).
5. Magnet seat: degrease, grit blast Sa 2½ just before bonding; no paint inside. Outside zinc spray + epoxy (C5).
6. Balance finished rotor (drum, magnets, sleeve, disc, cover) to ISO 21940-11 G2.5 at 250 rpm, correction on the rear flange ring.
7. Passive cooling (doc 23 §4a): 36 inner fins t3 S355, L-shaped, fillet-welded both sides to hub web and shell bore, every 10°, 9 mm clear of the end windings; 48 outer ribs 4 × 20 S355 on the shell outside, every 7.5°, intermittent fillet welds 30/60 both sides, ends rounded. Weld fins and ribs before stress relief and machining.
8. Rear flange ring welded to the shell full round, continuous: with the hub web it keeps the 9 mm shell round against the magnet pull (doc 23 §5). Seat roundness 0.05 after stress relief.

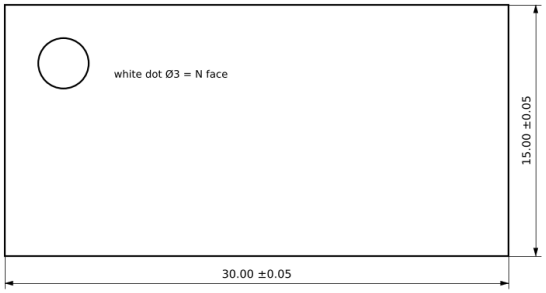
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

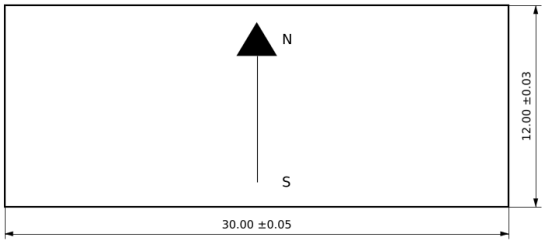
ROTOR DRUM WELDMENT (GENERATOR YOKE + BEARING HOUSING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-210C	C · 2026-11-05	1:3.8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N, welded, stress relieved	91.8 kg	Claude / not checked

VIEW A: N FACE (5:1)



VIEW B: END, magnetised through 12 (5:1)



Item	Requirement
Grade	N45SH: Br ≥ 1.32 T, HcJ ≥ 1,592 kA/m (20 kOe), (BH)max ≥ 342 kJ/m³ at 20 °C
Curves	J-H and B-H per batch at 20, 60, 100, 120 °C; αBr, βHcJ stated
Magnetisation	Through the 12 mm thickness, saturated; axis within 2° of the face normal
Tolerances	30 ±0.05, 15 ±0.05, 12 ±0.03; faces flat 0.02, parallel 0.02
Edges	Chamfer 0.2 to 0.3 × 45° all 12 edges; no chips > 0.3
Coating	NiCuNi 10 to 20 µm; 72 h neutral salt spray clean; adhesion OK with the epoxy
Uniformity	Moment of each piece within ±1.5 % of the lot mean; value per serial
Marking	White dot Ø3 on the N face; serial on the packaging position
Quantity	200 + 10 spare per generator; 8 extra for Q1 tests
Packing	Repelling layout in non-magnetic spacers; air freight per IATA PI 953

NOTES

1. One part for N and S poles: N poles face N towards the stator, S poles are the same block flipped (dot hidden against the seat).
2. Solid block (not two 6 mm pieces bonded). If a maker offers only 30 × 15 × 6, the doc 20 bonded pair is the fall-back with identical magnetics.
3. Five blocks per pole along the stack (15 mm pieces = axial segmentation for eddy loss, 4.7 W at rated).
4. Release of N45SH for production is conditional on the 100 °C design basis (doc 19 §16.3, doc 23 §4): bench magnet temperature ≤ 75 °C, temperature watch, no deliberate phase shorting.

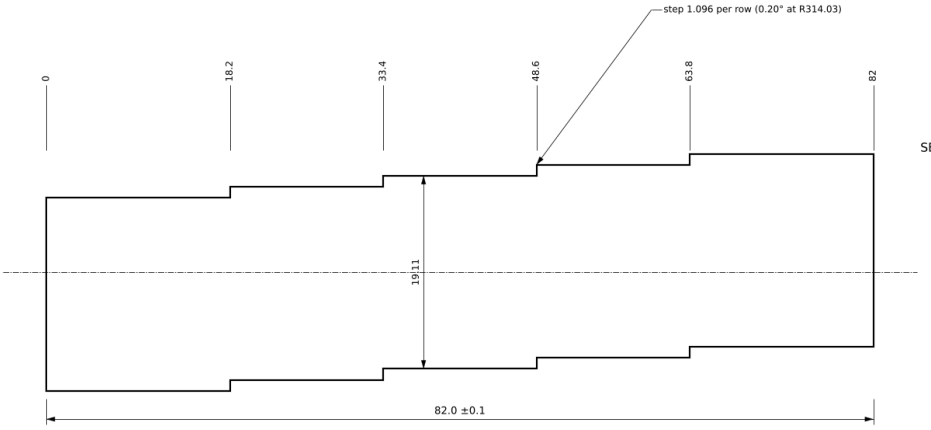
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

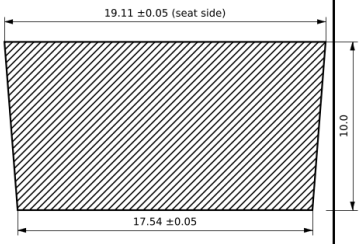
MAGNET BLOCK 30 × 15 × 12 (PURCHASE DRAWING)

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-220E	E · 2026-11-05	5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
Sintered NdFeB N45SH, NiCuNi	41.0 g each, 8.21 kg per generator	Claude / not checked

PLAN, seen from the stator (3:1). Steps set the row skew.



SECTION (5:1), flanks follow the block sides



NOTES

1. One bar per pole gap (40 per rotor + 4 spare). Bar sits on the seat (wide face), bonded with the magnet epoxy; ends engage 4 mm pockets in the end rings NP1-222C.
2. Row boundaries (ordinates from the front end): steps of 0.20° each; the bar is identical for every gap. Mark the front end.
3. Side faces are the datum for the blocks: 0.10 glue clearance each side (comb width = gap between neighbouring blocks − 0.20).
4. G11 (class H), ±0.05 on widths; edges deburred, no delamination. Prove on the Q1 panel (doc 18 §10).

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

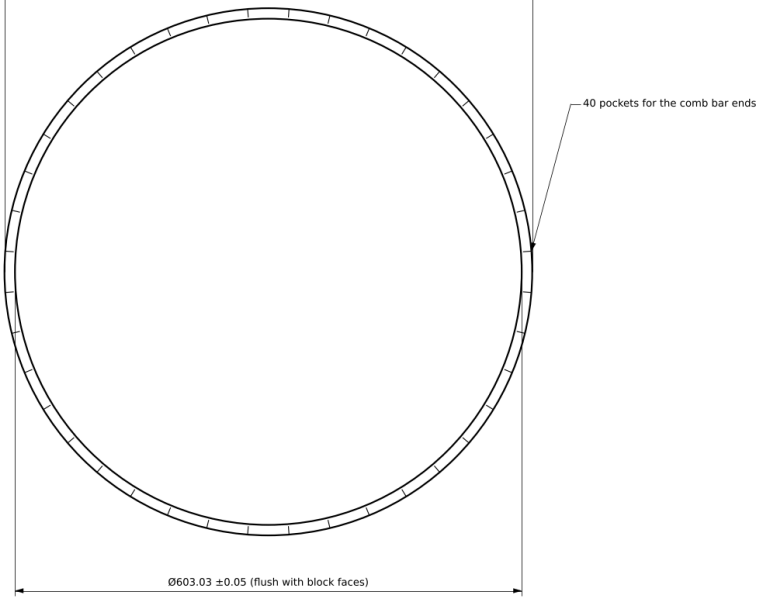
12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STEPPED POLE COMB BAR (40 + 4 PER ROTOR)

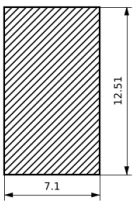
DRAWING NO.	REV / DATE	SCALE (A3)
NP1-221C	C · 2026-11-05	3:1 / 5:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy sheet, CNC machined	≈ 25 g	Claude / not checked

Ø628.05 −0.05/−0.10 (on the seat)

VIEW (1:4), front ring; rear ring mirrored



SECTION (4:1)



NOTES

1. Front ring at x 335 to 342.1, rear ring at x 417.9 to 425 (both 7.1 wide). They close the magnet pack and hold the comb bar ends.
2. Pockets: 19.11 wide (seat side), 10 deep radially, 4 deep axially, on the gap centre lines (4.5° + n × 9°), front ring rotated −0.40°, rear ring +0.40° (row skew).
3. May be made in 4 bonded arcs if no G11 tube of this size is available; joints on pocket centre lines.
4. Bonded to the seat with the magnet epoxy, then covered by the moulded glass sleeve.

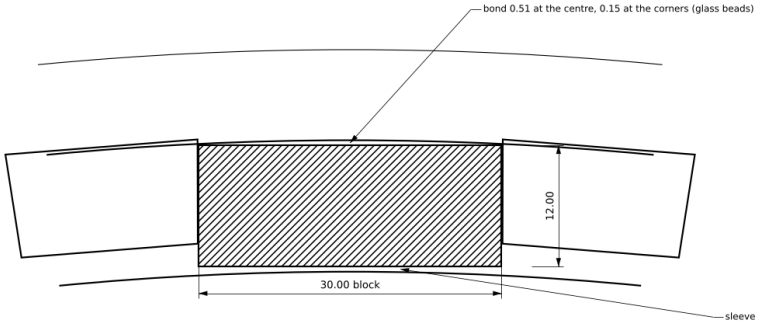
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

MAGNET END RINGS, FRONT AND REAR

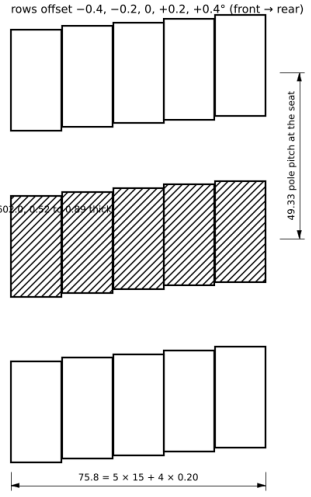
DRAWING NO.	REV / DATE	SCALE (A3)
NP1-222C	C · 2026-11-05	1:4 / 4:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
G11 glass-epoxy, machined from tube or segments	0.12 kg each	Claude / not checked

POLE SECTION (3:1), centre line = pole N1



LAYOUT, 3 POLES UNROLLED (1:1), front row left

rows offset −0.4, −0.2, 0, +0.2, +0.4° (front → rear)



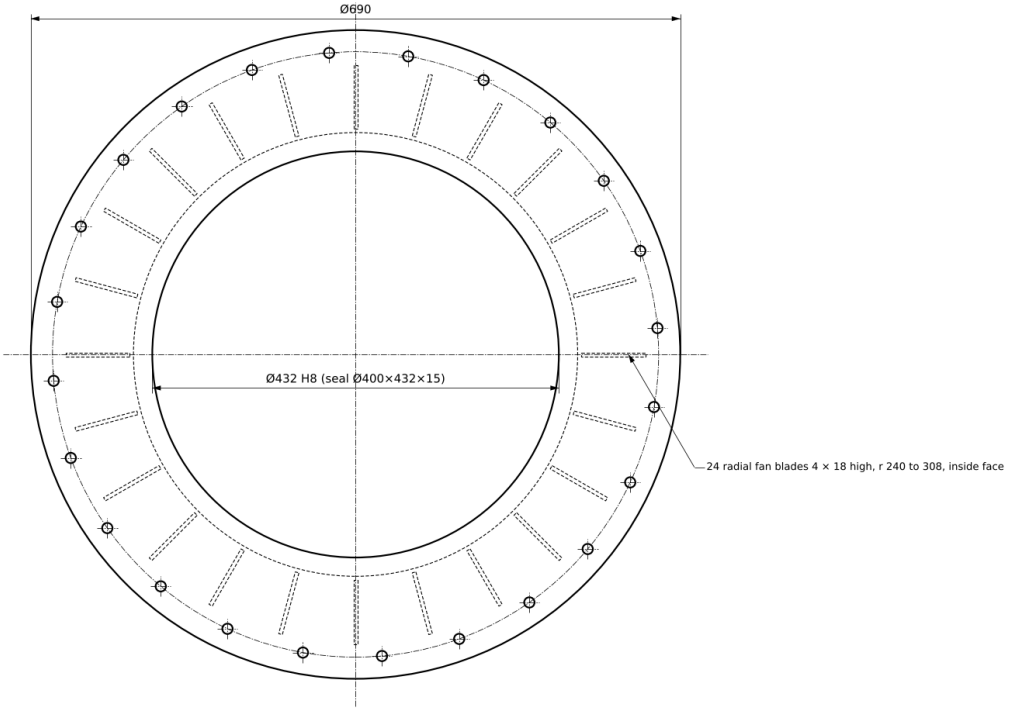
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Bond: two-component toughened epoxy, RT cure, 80 °C post-cure, with 0.15 glass beads (doc 18 §4.2). Blocks placed with jig NP1-T02 against the comb bar faces; all N poles first, then S.
2. Sequence per pole: comb bars and end rings bonded and cured first; blocks row by row from the front; check each row against the template (skew 0.20 ±0.02° per row).
3. Pole map NP1-T10 after cure: every pole centre within ±0.05° of nominal, flux per pole within ±1.5 %.
4. Sleeve: E-glass/epoxy moulded on the Al mandrel NP1-T06 located on both cup seats; bore Ø602.0 ±0.03, thickness 0.52 at the block centres, 0.89 at the edges; covers the end rings.
5. Q1 check (doc 18 §10): sleeve thickness by section 0.45 to 0.95; air gap feeler check 2.0 ±0.1 all round after joining.

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)		
ROTOR MAGNET PACK: POLE SECTION, LAYOUT AND SLEEVE		
DRAWING NO. NP1-223C	REV / DATE C · 2026-11-05	SCALE (A3) 3:1 / 1:1
MATERIAL assembly	MASS (CALC.) 8.2 kg magnets	DRAWN / CHECKED Claude / not checked

VIEW FROM FRONT (inside), plate 10, seal ring 15 × 20 at x 450 to 465



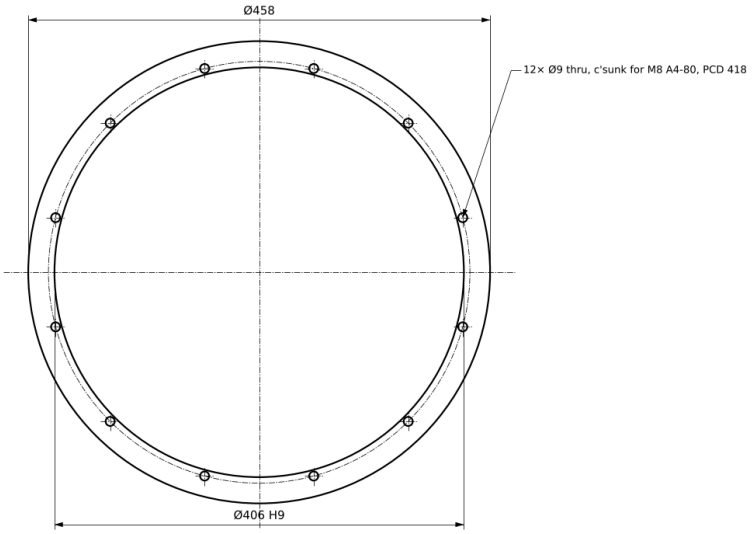
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Rev C (doc 23 §6): 24 radial blades welded (TIG) or bolted to the inside face; they stir the internal air over the end windings and carry heat to the cover ($h_{int} \approx 40$ W/m²K instead of 15, to be measured on the bench, doc 15 T5).
2. Blades span x 447 to 465: 8 mm clear of the potted end turns (x ≤ 439) and of the carrier rear stop (x 440); tips r 308 run 4 mm inside the shell bore Ø624.
3. Rear seal: radial lip seal Ø400×432×15 HNBR −40 °C on the spindle flange OD; breather port M24×1.5 with desiccant breather.
4. Balance with the rotor (NP1-210C note 6). Inspection port Ø60 with cover at 180° (not shown).

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)		
DRUM REAR COVER WITH SEAL CARRIER AND FAN BLADES		
DRAWING NO. NP1-212C	REV / DATE C · 2026-11-05	SCALE (A3) 1:3.6
MATERIAL EN AW-5083-H111 plate, ring and blades	MASS (CALC.) ≈ 9.0 kg	DRAWN / CHECKED Claude / not checked

VIEW (thickness 8.0, faces parallel 0.05)



NOTES
1. Laser cut from 8 mm plate, faces machined flat 0.05.
2. Must not touch the coils: inner end-winding radius ≥ 232; ring OD 229.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · GENERATOR REV C (DOC 23)

STATOR FRONT CLAMP RING

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-202	B · 2026-11-05	1:3.3
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
1.4404 (non-magnetic)	2.2 kg	Claude / not checked

10. Changes to other documents, questions and suggestions

Changes for building (against doc 18):

Doc 18 step	Rev C
§4.3 magnets	NP1-220E: N45SH 30 × 15 × 12, 210 pieces
§6.2 drum machining	Seat Ø628.05; full-round end welds; seat roundness 0.05
§6.4 comb	Stepped G11 bars NP1-221C, end rings NP1-222C
§6.5 bonding	One block at a time, row by row, no pair bonding
§6.8 sleeve	Sleeve 0.52 to 0.89 (Q1 checks the full range)
§6.1 drum weldment	36 inner fins and 48 outer ribs welded before stress relief (NP1-210C note 7)
§6.8 complete the rotor	Rear cover NP1-212C with 24 fan blades; balance with it fitted
§5 stator	Unchanged, 28 turns
Controller	Magnet temperature: derate 85 °C, stop 95 °C (doc 19 §16.3)

Questions for Martin:

1. ~~Outer ribs fit~~: Ø676 over the ribs accepted by Martin (2026-11-05).
2. **Magnet supplier:** 30 × 15 × 12 N45SH is less common than 30 × 15 × 6. Should I prepare a quote request for Neo Narva, VAC and two Chinese makers with this drawing?

Suggestions:

- **Q1 test panel with the real blocks:** bond 10 blocks on a seat sector with the stepped comb, cure, then a push-off test at 120 °C. This proves the single-block bond before the full rotor.
- **Bench test of temperatures** (doc 15): thermocouples on the end windings, slot centre, inside air and drum, at rated and 110 %. This calibrates the inside-air coefficient, the largest uncertainty in §4a.
- **3D end-effect check** of the five short rows (open from doc 20 O5).

Appendix. How to rerun

```
cd analysis/generator
python3 sh30.py S1x              # FEA of the Rev C pole, ~40 min -> results/sh30_S1x.json
python3 checks_rc.py             # thermal, mechanical, voltage -> results/rc_checks.json
python3 passive_cooling.py       # passive cooling options -> results/passive_cooling.json
cd ../../cad/generator_rc
python3 build_rc.py --png        # drawing set: PDF, DXF, PNG
python3 iso_rc.py --png          # isometric views IR01 to IR06
```